

## HCR Project 2026 - Week 2: Introduction to Programming Languages

Mentor - Prof. Dr. Mohammed Aquil Mirza, COMP

Student Mentors - Mr. Ruan Haozhe, Mr. Jiang Suyu

The Hong Kong Polytechnic University (PolyU), Hong Kong

**Deadline: 29:59, 31 May 2026 (5:59 AM, 1 Jun2026)**

Name: \_\_\_\_\_ School: \_\_\_\_\_

1. Write down the expected output of `rustc -version` after successful installation of Rust.

-----[1]-----

2. Modify the `tick` function in the `haircutting_web`, `src/script.rs` file. So that after the modification, the cursor will move linearly.

```
fn tick(&mut self, t: u64, sim: &mut Simulation) {
    // Phase 1
    const APPROACH: u64 = 42;
    if t < APPROACH {
        let _ = sim.apply_command(RuntimeCommand::Clipper(
    ↪ ClipperCommand::MoveForward));
        return;
    }

    // Phase 2
    if t == APPROACH {
        let _ = -----[2]-----,
    }

    // Phase 3: sweep the clipper from x = -1.0 to x = +1.0 over
    ↪ 200 ticks.
    const SWEEP_TICKS: f32 = 200.0;
    let progress = ((t - APPROACH) as f32 / SWEEP_TICKS).clamp
    ↪ (0.0, 1.0);
    let x = -----[3]-----;
    let z = -----[4]-----;
    let _ = sim.apply_command(RuntimeCommand::Clipper(
    ↪ ClipperCommand::SetTargetXz {
        x,
        z,
    }));
}
```

|     |  |
|-----|--|
| [1] |  |
| [2] |  |
| [3] |  |
| [4] |  |